



1 Introduction

You will write five assembly functions that operate on a library of books. Each function is called from a C program, so it must follow the calling convention.

The Book struct

```
struct Book { int id; double rating; int pages; };
```

Your assembly must use the same memory layout C uses for this struct. You are not given a definition for assembly.

2 Building and testing

A makefile and a sample harness `main.c` are provided:

```
make          # builds ./library
./library     # runs your functions on the sample data
```

The marker uses different data, so a correct run of `./library` does not guarantee marks.

3 Submission and grading

Submit `task1.asm` to `task5.asm`, one function per file, each made `global`. Each task is marked on its own: the marker links your file with a C harness, runs it, and compares the printed result byte for byte. Do not print anything yourself; return the result.

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4 The functions

`n` is the number of books and is at least 1.

Task 1: `total_pages` (4 marks)

```
long total_pages(Book* books, long n);
```

Return the sum of the `pages` field over the books.

Task 2: `average_rating` (6 marks)

```
double average_rating(Book* books, long n);
```

Return the mean of the `rating` field.

Task 3: `count_above` (6 marks)

```
long count_above(Book* books, long n, double threshold);
```

Return the number of books whose `rating` is greater than `threshold`.

Task 4: `best_book` (6 marks)

```
Book* best_book(Book* books, long n);
```

Return a pointer to the book with the highest `rating`. If several books share the highest `rating`, return the first of them.

Task 5: `weighted_rating` (8 marks)

```
double weighted_rating(Book* books, long n);
```

Return

$$\frac{\sum_i \text{rating}_i \times \text{pages}_i}{\sum_i \text{pages}_i}.$$

5 Mark distribution

Task	Mark
Task 1: total_pages	4
Task 2: average_rating	6
Task 3: count_above	6
Task 4: best_book	6
Task 5: weighted_rating	8
Total	30